

# TECHNOLOGY



## Designing Microsoft Azure Infrastructure Solutions: AZ:305

## Design for Cost Optimization





# Learning Objectives

By the end of this lesson, you will be able to:

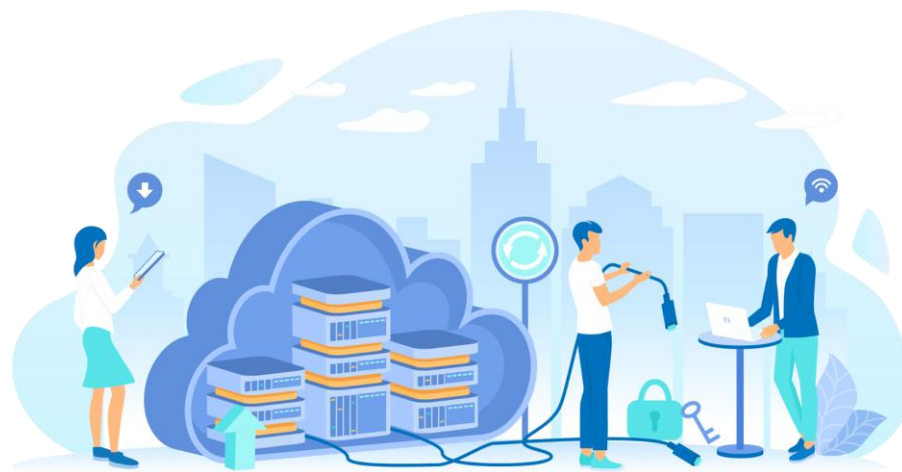
- Develop a solution to optimize expenditure and resource allocation for efficient cost management
- Design a solution to implement robust cloud cost management tools and analytics for detailed and actionable cost reporting
- Develop solutions to minimize costs through resource optimization, strategic purchasing, and efficient scaling
- List the key principles for cost management for efficient resource use and financial optimization in cloud environments



## Recommend a Solution for Cost Management and Cost Reporting

# Introduction

In the current landscape, businesses operate within the following sophisticated cloud environment:



Managing a vast array of cloud resources

Microsoft Azure



Utilizing multiple cloud platforms concurrently



Operating across various business units and projects

# Azure Cost Management and Billing

It is a set of tools that allows the user to evaluate, manage, and optimize the costs of the workload.



- Users only pay for what they use with Azure products and services.
- Without effective analysis and monitoring, workload expenses can skyrocket due to the ease with which businesses can deploy more resources.

# Azure Cost Management

It displays organizational cost and usage patterns. It shows usage-based costs consumed by Azure services and third-party marketplace offerings.



Organizations can:

- Conduct cost analysis
- Create a budget
- Review recommendations
- Export the data





# Azure Cost Management

Organizations must address the cost management issues before investing in cloud computing resources. To successfully implement cost management and optimize costs, the organization must:



- Ensure they have the right tools for success
- Be accountable for the expenses
- Develop a strategy to optimize spending



# Optimize with Azure Cost Management

Organizations must align the following groups to manage costs successfully:

Finance

Managers

App teams



- Users approve budget requests across the organization based on cloud spending forecasts.
- Users pay the corresponding bill and assign costs to other teams to drive accountability.

# Optimize with Azure Cost Management

Organizations must align the following groups to manage costs successfully:

Finance

Managers

App teams



Organizational decision-makers must comprehend cloud expenses to maximize their returns and optimize cost efficiency.

# Optimize with Azure Cost Management

Organizations must align the following groups to manage costs successfully:

Finance

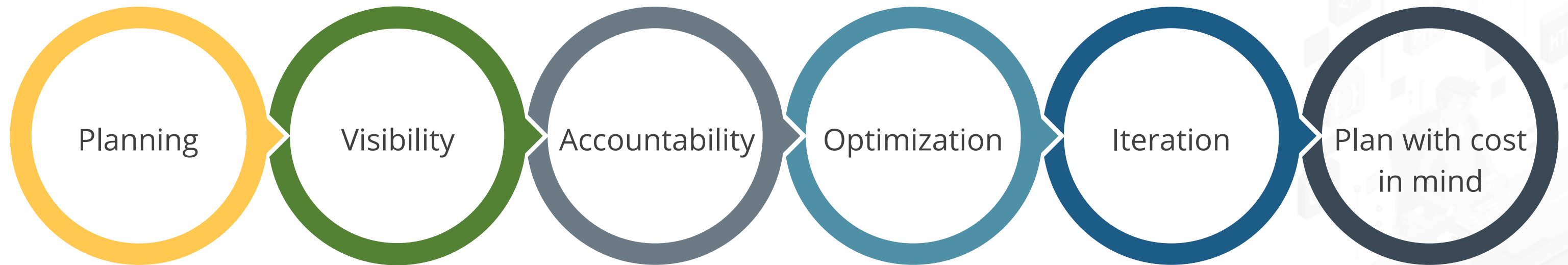
Managers

App teams



- Engineers who handle cloud resources regularly create services to meet organizational needs.
- App teams require flexibility to deliver the maximum value within their budget constraints.

# Cost Design Principles





# Planning

Comprehensive upfront planning tailors cloud usage to specific business requirements.

The answers to the questions below help select the offerings right for you. You determine the infrastructure to use and how it's used to maximize your Azure efficiency.

- For which business problem is a solution sought?
- Which usage patterns are expected from the resources?
- What are the expected workload size and usage patterns?
- Can resources be shut down during off-peak hours?
- What is the data retention period?
- Which Azure regions offer the best balance of cost and performance?
- Can existing software licenses be reused?

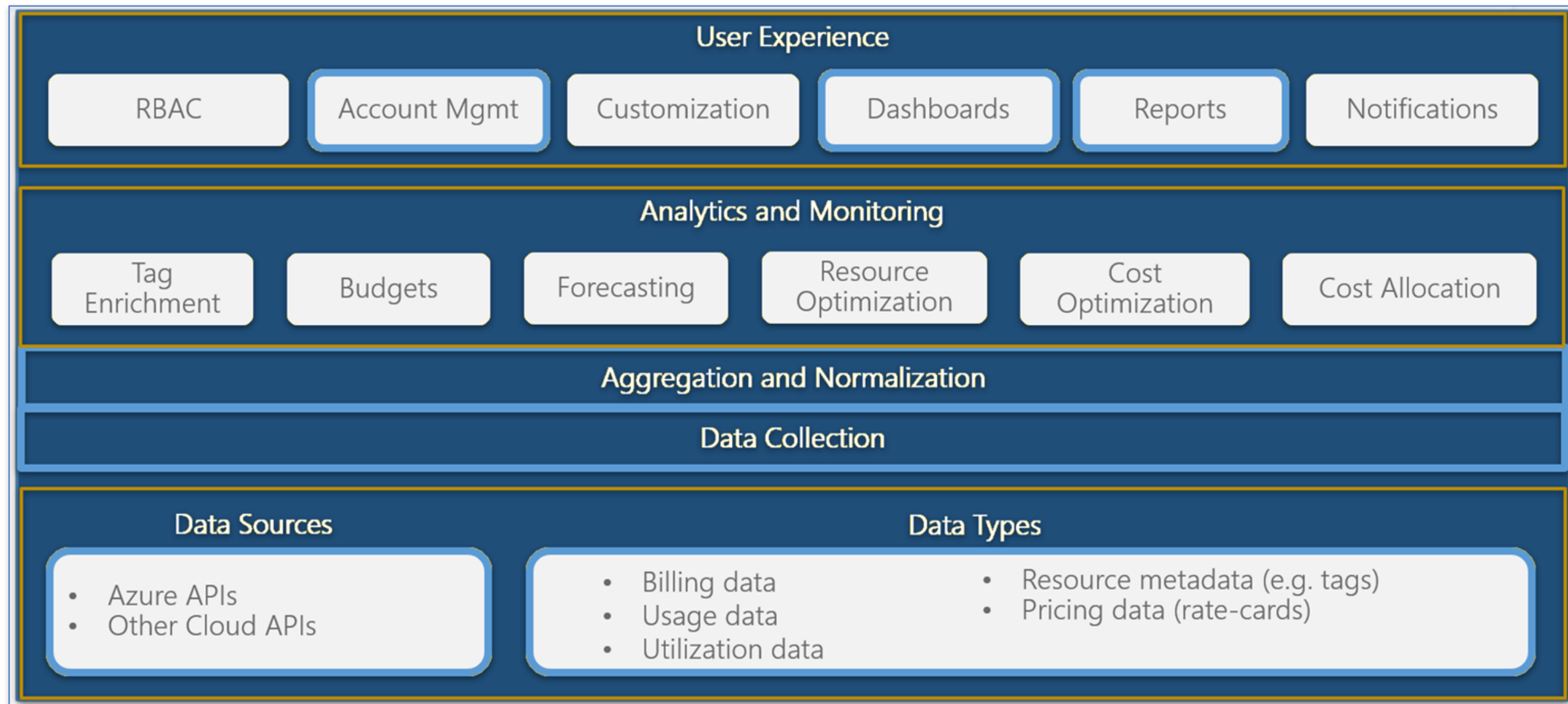
# Visibility

Cost management, when properly structured, assists organizations in informing people about the Azure costs for which they are responsible, as well as the money they spend.



# Visibility

Azure provides tools to help you identify underused resources, remove waste, and maximize cost-saving opportunities.



# Accountability

Organizations can implement accountability by:



- Extending visibility to stakeholders
- Making stakeholders aware of the costs
- Making stakeholders accountable
- Setting clear budgets
- Setting alerts (be notified of any anomalies)



# Optimization

It means taking measures to reduce costs. Organizations should make the most of it based on the information acquired through cost visibility and planning.



**Optimization is not a one-time effort.**

Cloud is a constantly changing place with new projects, new applications, and changing loads.

# Optimization

It can be of two types:

## Resource optimization

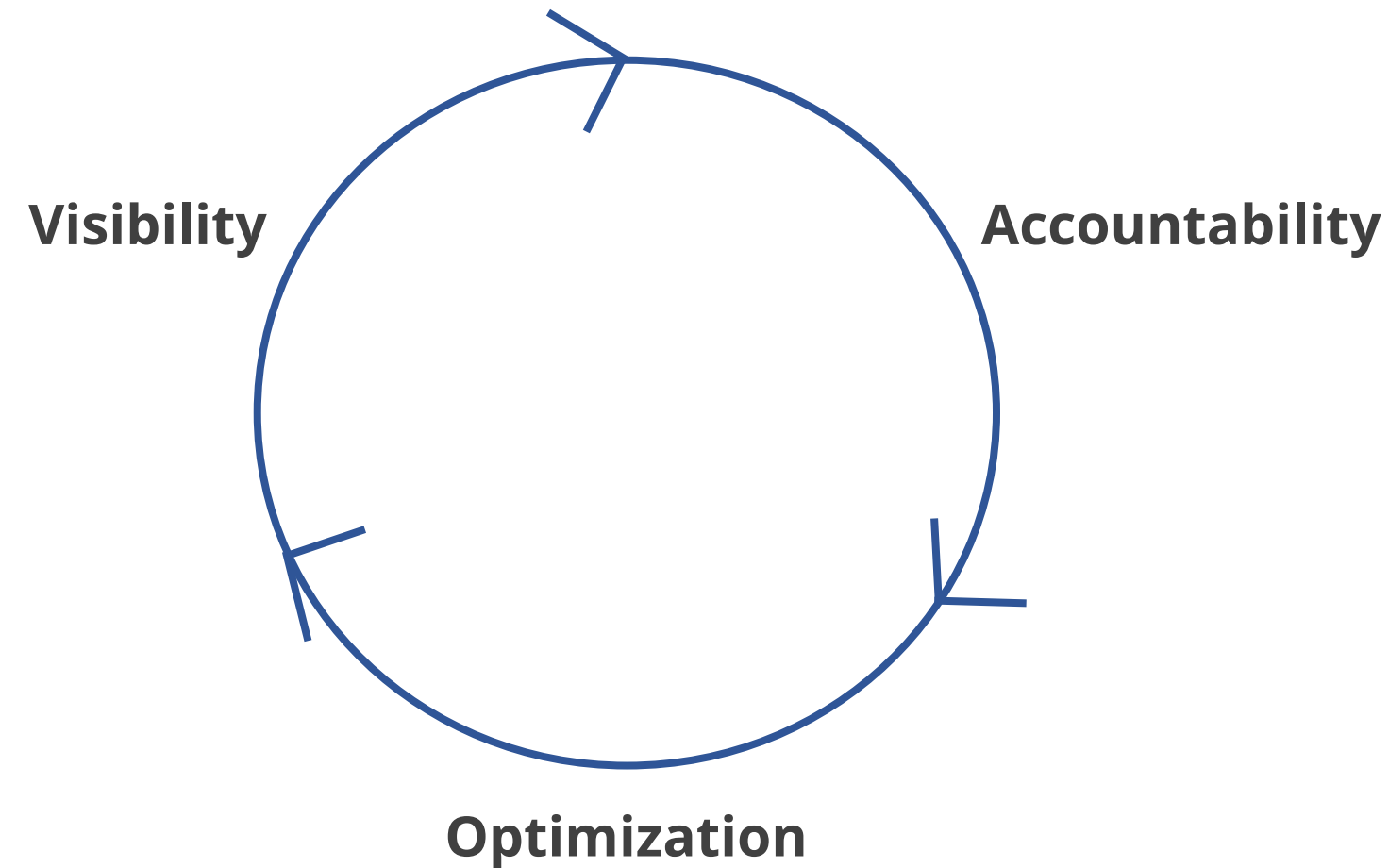
- Correct families and sizes
- Correct resource or service types

## Cost optimization

- When to buy more or less reserved instances

# Iteration

Ensure everyone in the organization actively participates in the ongoing cost management lifecycle, making it a key tenet of responsible cloud governance.



# Design with Cost in Mind

The common questions for cost analysis include:



- Cost estimation for the current month
- Anomaly investigation
- Invoice reconciliation
- Internal chargeback

## Recommend Solutions to Minimize Costs



# Plan and Estimate Azure Cost

Before deploying cloud resources, organizations must assess the following items:



The Azure offer that best meets user needs



The estimation of their cost



The resources users plan to use

Azure's assessment tools help determine the investment needed for the workloads, allowing users to choose the optimal configuration.

# View Cost Breakdown by Azure Service

1

Review invoiced charges in Cost Analysis

Scope : Contoso (Demo)

VIEW Invoice details

Mar 2020

+ Add filter

ACTUAL COST (USD)

FORECAST UNAVAILABLE

BUDGET: DEV\_ENV

\$91,412.07

--

\$100 /mo

Group by: Meter

Granularity: None

Table

Filter items

490 rows

| Publisher type | Charge type | Service name              | Service tier                       | Meter                          | Part Number | Cost        |
|----------------|-------------|---------------------------|------------------------------------|--------------------------------|-------------|-------------|
| azure          | usage       | virtual machines          | fsv2 series windows vm             | f72s v2                        | aad-10036   | \$11,634.61 |
| azure          | usage       | azure stack hub           | azure stack-windows vm             | 1 core                         | aaa-56070   | \$9,432.73  |
| azure          | usage       | hdinsight                 | hdinsight d series                 | d4                             | n7h-01995   | \$7,455.47  |
| azure          | usage       | expressroute              | expressroute                       | premium metered data 10 gbp... | j2q-00792   | \$6,398.39  |
| azure          | usage       | virtual machines licenses | sql server ent                     | 16 vcpu vm license             | n7h-07116   | \$3,936.78  |
| azure          | usage       | virtual machines          | dv2 series windows vm              | d15 v2/ds15 v2                 | aaa-14527   | \$2,926.18  |
| azure          | usage       | sql database              | sql db single/elastic pool gen ... | vcore                          | aad-17849   | \$2,832.64  |

Note: The above screenshot is for your reference. The interface is subject to change per Azure updates. You can login to your account to check your Cost Analysis dashboard.

# View Cost Breakdown by Azure Service

2

View cost breakdown by Azure resource

Scope : Contoso IT - demo

\* Cost by resource

Home

Built-in views

Accumulated costs

Cost by resource

Daily costs

Cost by service

Invoice details

Oct 2019

Add filter

ACTUAL COST (USD) ⓘ

\$5,638.76

BUDGET: DEVTESTSPENDLIMIT

\$13,000 /mo

Group by: Resource

Granularity: None

Table

| Resource                    | Resource group            | Location         | Tags                        | Cost                       |            |
|-----------------------------|---------------------------|------------------|-----------------------------|----------------------------|------------|
| hybrid-k8s-azmon            | Log Analytics workspace   | us east          | hybrid-k8s-azmon-rg         | businessowner, compos...   | \$1,704.04 |
| contoso-retail-it           | Log Analytics workspace   | us east          | contosoazurehq              | businessowner, compos...   | \$615.93   |
| yqiducontoso-eus-arcache    | Storage account           | us south central | contoso-recovery-vault-seus | environment:prod           | \$311.29   |
| defaultworkspace-e4272...   | Log Analytics workspace   | us east          | defaultresourcegroup-eus    | businessowner, compos...   | \$295.65   |
| contoso-jumpbox-az-firewall | Firewall                  | us south central | contoso-jumpbox-firewall    | Not applicable             | \$160.15   |
| chqdisksk7owxak6tjx7u       | Storage account           | us south central | contosoazurehq              | displayname:vmstorage      | \$107.51   |
| vm-scale-set                | Virtual machine scale set | us east          | contoso-chef-automate-vms   | x-application:rhel-hard... | \$78.71    |

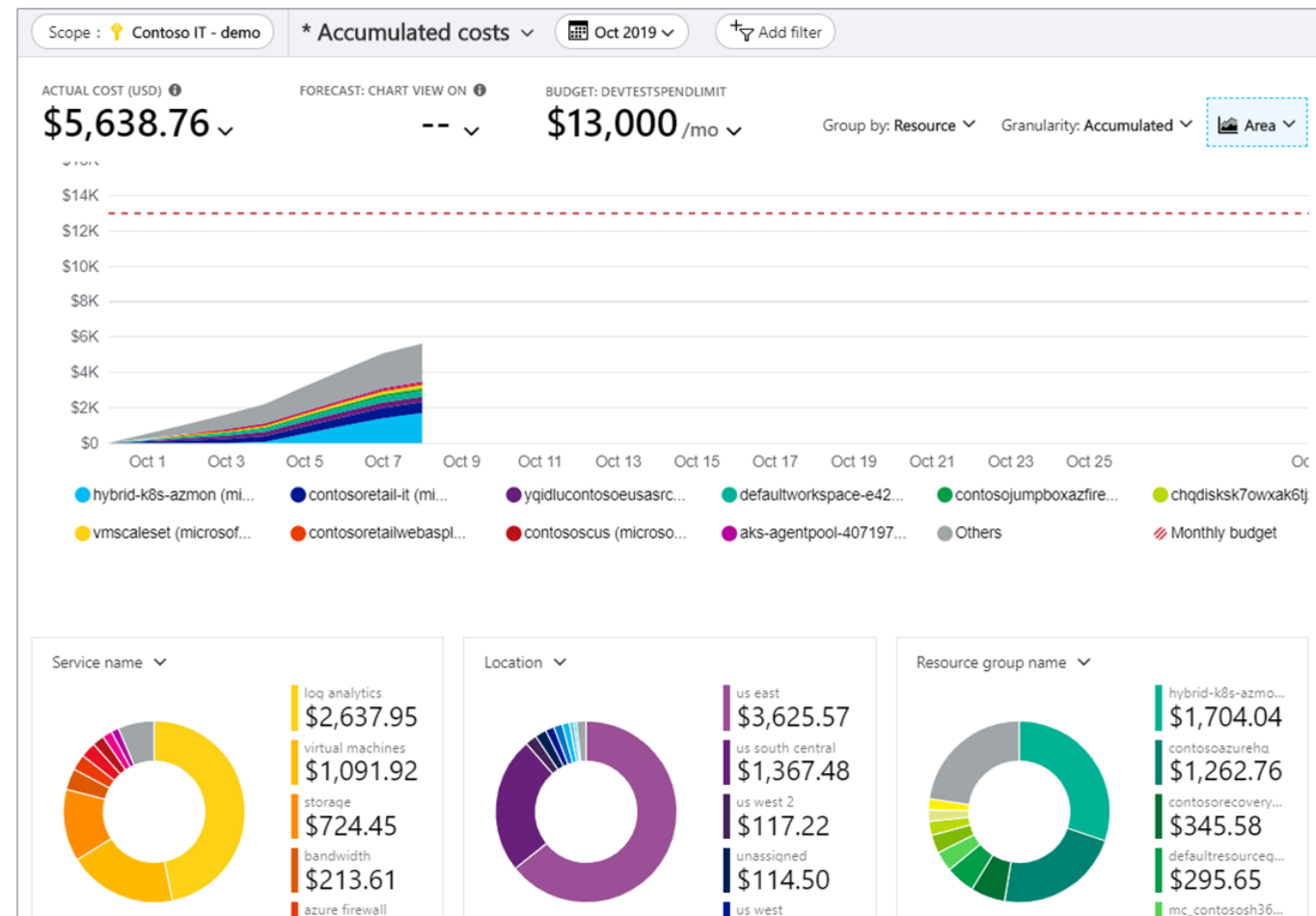
- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Select **Cost by resource**
- Change the view to **Table**

Note: The above screenshot is for your reference. The interface is subject to change per Azure updates. You can login to your account to check your Cost Analysis dashboard.

# View Cost Breakdown by Azure Service

3

View cost breakdown by selected dimensions

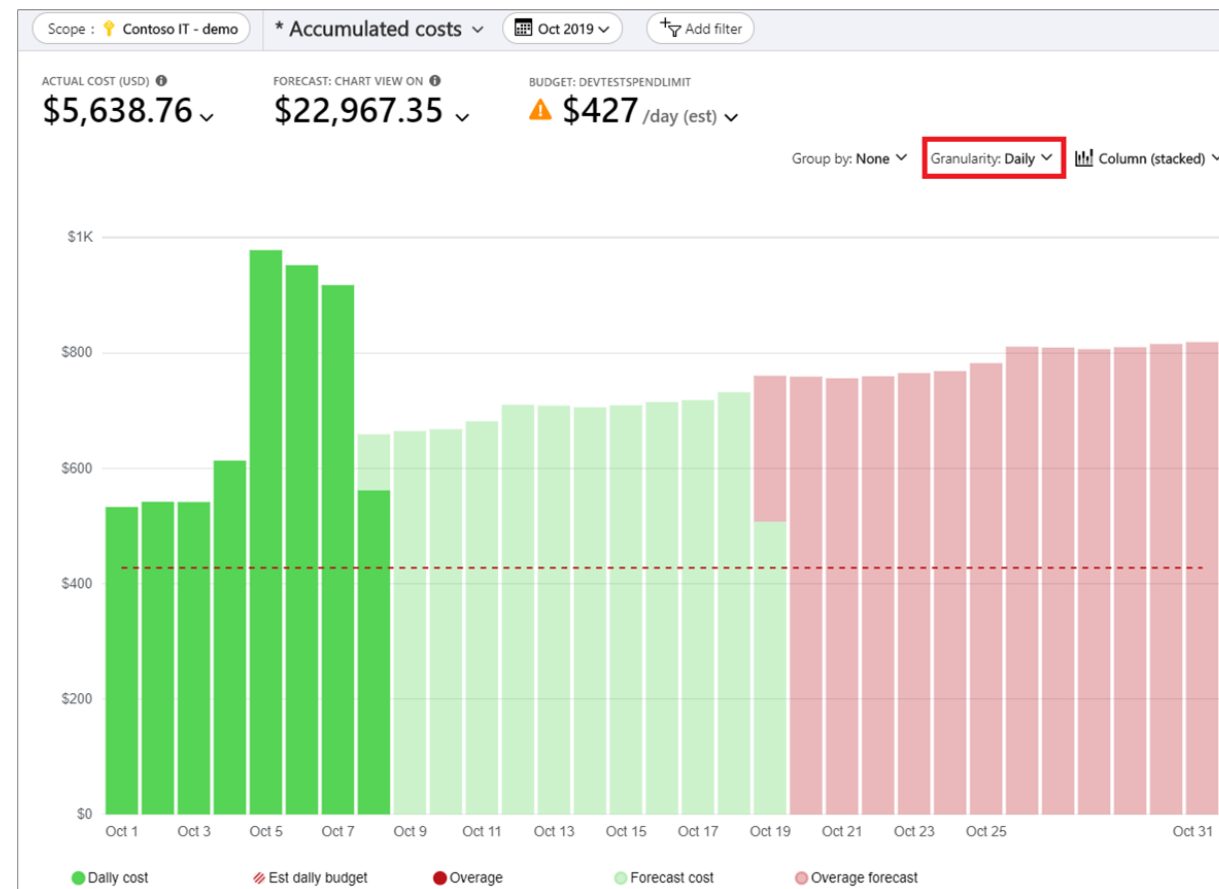


- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Select the **Group by filter**

Note: The above screenshot is for your reference. The interface is subject to change per Azure updates. You can login to your account to check your Cost Analysis dashboard.

# View Costs by Day or Month

Analyzing daily and monthly costs helps identify peak periods, allowing a better understanding of when expenses are higher.



- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Set the **Granularity** to **Monthly** or **Daily**

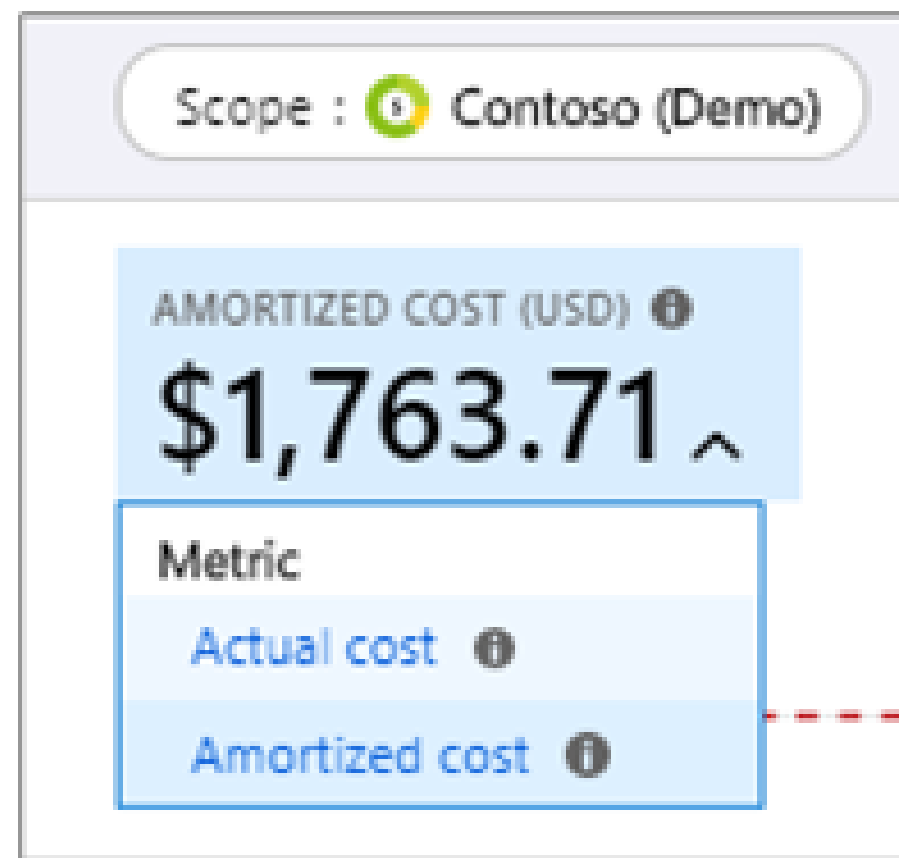
Note: The above screenshot is for your reference. The interface is subject to change per Azure updates. You can login to your account to check your Cost Analysis dashboard.

# View Reservation Charges

The reservation charges are shown as an actual or an amortized cost.

## Actual cost

Shows the purchase as it appears on the bill



## Amortized cost

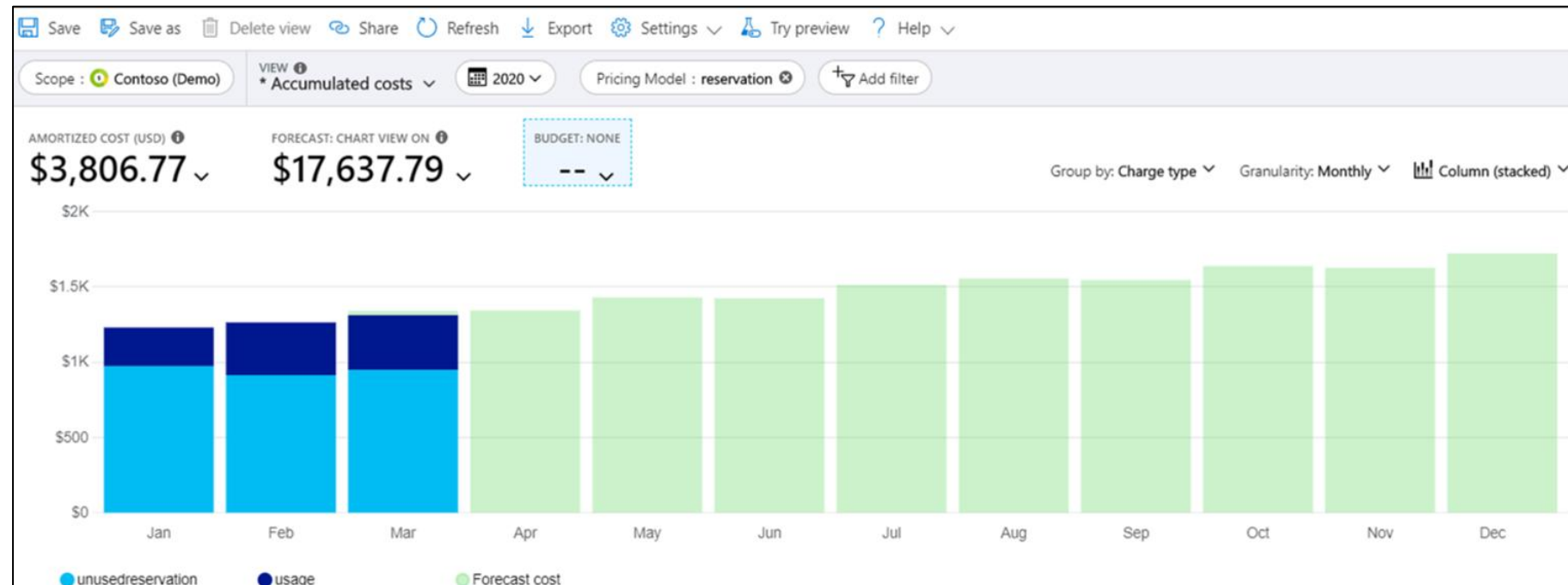
Shows a reservation purchase split as an amortized cost over the duration of the reservation term

**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.



# View Reservation Charges

Identify the current monthly waste in the reservation purchase costs



**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.

- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Add a filter for the **Pricing Model: Reservation**
- Select the **Amortized Cost** view
- Set the granularity to **Monthly**
- Set the period to the current year or your reservation term
- Set the chart type to **Column (stacked)**
- Group charges by **Charge Type**
- Review the results for unused reservation values

# View Cost for a Specific Tag

Tag support is for usage reported after the tag has been applied to the resource.

The screenshot displays the Azure Cost Management + Billing interface. At the top, the scope is set to 'Contoso IT - demo' and the view is 'Accumulated costs' for 'Oct 2019'. The actual cost is \$5,638.76, and the budget is \$13,000/mo. A 'Group by' dropdown menu is open, showing various options like 'Publisher type', 'Reservation', 'Resource', etc. The 'Tag' option is highlighted at the bottom of the list. The table below shows three rows of data with 'TagValue' as 'Untagged', 'prod', and 'test'.

| TagValue |
|----------|
| Untagged |
| prod     |
| test     |

- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Select **Group by** for the tag

**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.

# Download Usage Details

In the Azure portal, navigate to the **Usage and Charges** tab for a billing account or subscription

| Month                                                                                                                                                         | Charges against credits | Service overage | Billed separately | Azure marketplace | Total charges | Download |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------|-------------------|-------------------|---------------|----------|
| Oct 2019                                                                                                                                                      | 0.00                    | 18,807.27       | 0.00              | 4.95              | 18,812.22     | ↓        |
| Sep 2019                                                                                                                                                      | 0.00                    | 72,273.21       | 0.00              | 0.00              | 72,273.21     | ↓        |
| Aug 2019                                                                                                                                                      | 0.00                    | 72,143.63       | 0.00              | 0.00              | 72,143.63     | ↓        |
| Jul 2019                                                                                                                                                      | 0.00                    | 69,136.60       | 0.00              | 0.00              | 69,136.60     | ↓        |
| Jun 2019                                                                                                                                                      | 0.00                    | 72,232.05       | 0.00              | 0.00              | 72,232.05     | ↓        |
| May 2019                                                                                                                                                      | 0.00                    | 89,889.50       | 0.00              | 396.00            | 90,285.50     | ↓        |
| Apr 2019                                                                                                                                                      | 0.00                    | 77,323.82       | 0.00              | 1,248.91          | 78,572.73     | ↓        |
| Mar 2019                                                                                                                                                      | 0.00                    | 71,091.47       | 0.00              | 3,877.99          | 74,969.46     | ↓        |
| Feb 2019                                                                                                                                                      | 0.00                    | 66,806.93       | 0.00              | 2,565.65          | 69,372.58     | ↓        |
| Jan 2019                                                                                                                                                      | 0.00                    | 74,920.08       | 0.00              | 3,313.46          | 78,233.54     | ↓        |
| Dec 2018                                                                                                                                                      | 0.00                    | 97,609.51       | 0.00              | 2,731.18          | 100,340.69    | ↓        |
| Nov 2018                                                                                                                                                      | 0.00                    | 107,054.43      | 172.71            | 4,746.55          | 111,973.69    | ↓        |
| <b>Total</b>                                                                                                                                                  |                         |                 |                   |                   |               |          |
| Charges against credits: 0 USD   Service overage: 889.29 K USD   Billed separately: 172.71 USD   Azure marketplace: 18.88 K USD   Total charges: 908.35 K USD |                         |                 |                   |                   |               |          |

- Navigate to **Cost Management + Billing > Billing > Usage + charges**
- Select the line item to download from and then select the download symbol
- Select the usage file to download

**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.

# View Monthly EA Cost Breakdown

Understanding the accrual and invoicing of costs over time enables engaging stakeholders for responsible cost management.

Timespan

Last 12 months

\*Currency: USD

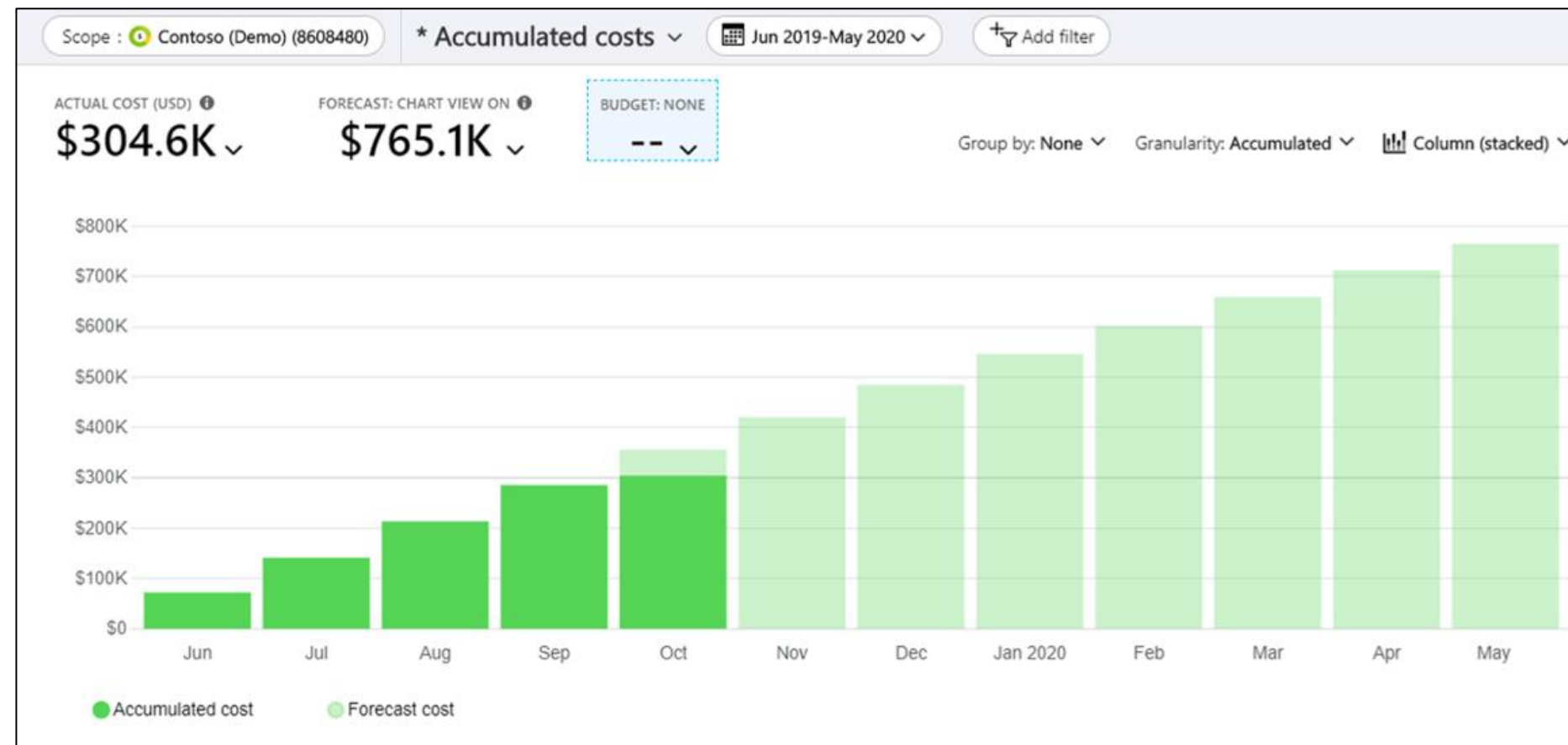
| Month                                                                                                                                                                                                                                                                                                            | Charges against credits | Service overage | Billed separately | Azure marketplace | Total charges | Download |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------|-------------------|-------------------|---------------|----------|
| Oct 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 18,807.27       | 0.00              | 4.95              | 18,812.22     | ↓        |
| Sep 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 72,273.21       | 0.00              | 0.00              | 72,273.21     | ↓        |
| Aug 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 72,143.63       | 0.00              | 0.00              | 72,143.63     | ↓        |
| Jul 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 69,136.60       | 0.00              | 0.00              | 69,136.60     | ↓        |
| Jun 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 72,232.05       | 0.00              | 0.00              | 72,232.05     | ↓        |
| May 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 89,889.50       | 0.00              | 396.00            | 90,285.50     | ↓        |
| Apr 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 77,323.82       | 0.00              | 1,248.91          | 78,572.73     | ↓        |
| Mar 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 71,091.47       | 0.00              | 3,877.99          | 74,969.46     | ↓        |
| Feb 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 66,806.93       | 0.00              | 2,565.65          | 69,372.58     | ↓        |
| Jan 2019                                                                                                                                                                                                                                                                                                         | 0.00                    | 74,920.08       | 0.00              | 3,313.46          | 78,233.54     | ↓        |
| Dec 2018                                                                                                                                                                                                                                                                                                         | 0.00                    | 97,609.51       | 0.00              | 2,731.18          | 100,340.69    | ↓        |
| Nov 2018                                                                                                                                                                                                                                                                                                         | 0.00                    | 107,054.43      | 172.71            | 4,746.55          | 111,973.69    | ↓        |
| Total                                                                                                                                                                                                                                                                                                            |                         |                 |                   |                   |               |          |
| <div><div>Charges against credits</div><div>0 USD</div></div> <div><div>Service overage</div><div>889.29 K USD</div></div> <div><div>Billed separately</div><div>172.71 USD</div></div> <div><div>Azure marketplace</div><div>18.88 K USD</div></div> <div><div>Total charges</div><div>908.35 K USD</div></div> |                         |                 |                   |                   |               |          |

- Navigate to **Cost Management + Billing > Overview**
- Select **Breakdown**
- Select the **Usage and Charges** tab and view the prior month's breakdown in the chosen timespan

**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.

# View Monthly EA Cost Breakdown

Use the graphical view of the enrollment's monthly costs to grasp cost trends and invoiced amounts for a specific period:



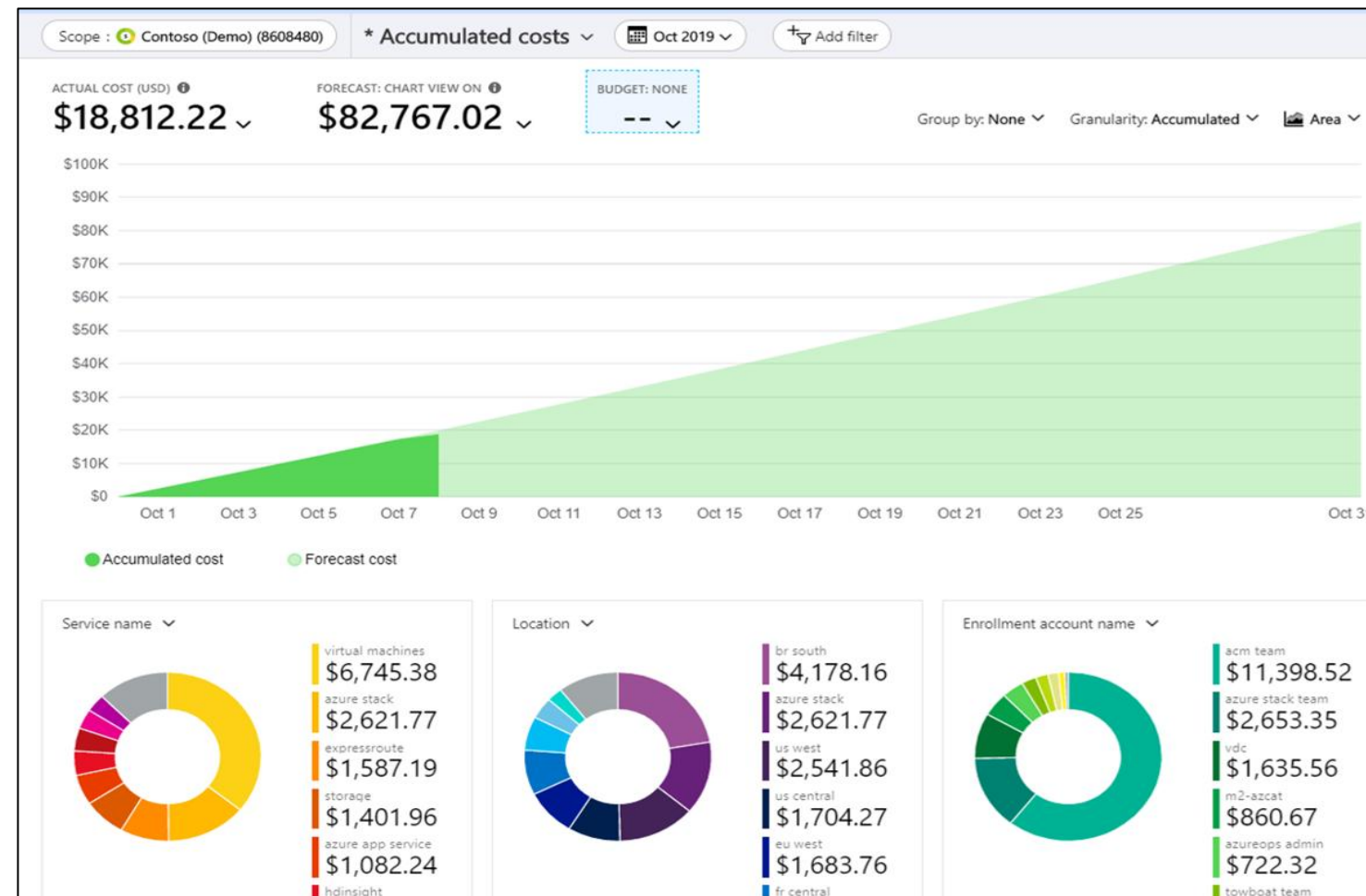
- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Select enrollment and set the enrollment term
- Set the granularity to **Monthly** and then set the view to **Column (stacked)**

**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.



# View Monthly EA Cost Breakdown

View the net accumulated charges over time to comprehend your organization's overall expenditures for a specific period.



- Navigate to **Cost Management + Billing > Cost Management > Cost analysis**
- Select the enrollment and then view the current accumulated costs

**Note:** The above screenshot is for reference. The interface is subject to change per Azure updates. Users can login to the account to check the Cost Analysis dashboard.



# Checklist: Design for Cost

Cost model

Capture clear requirements

Estimate the initial cost

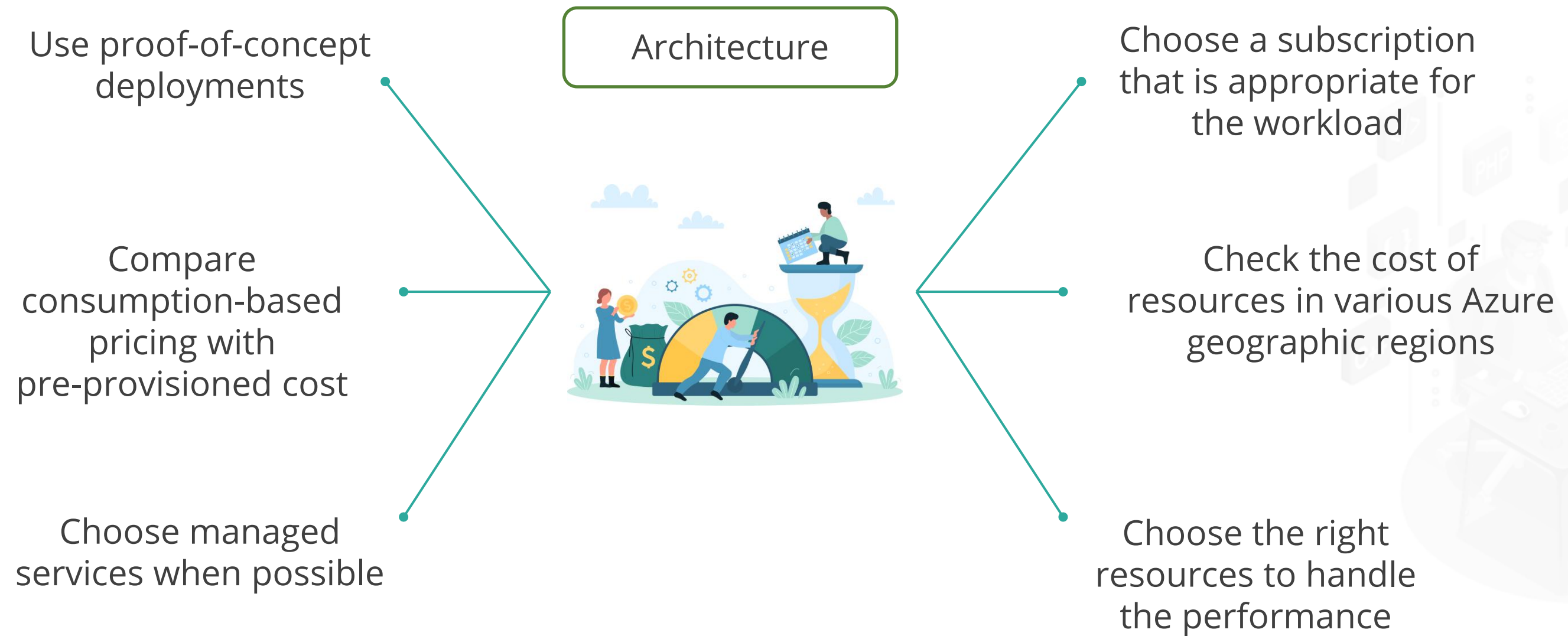
Define policies for the cost constraints defined by the organization

Identify shared assets

Plan a governance strategy



# Checklist: Design for Cost



# Checklist: Optimize Cost

Use the following checklist to optimize the workload:

Review underutilized resources

Use discount prices

Act continuously on the cost reviews

Have a scale-in and scale-out policy

Use reserved instances on long-running workloads

Reevaluate design choices

# Checklist: Optimize Cost

Use the following checklist to optimize the workload:

Choose the right storage tier

Optimize data transfer

Choose the right data store

Reduce load on servers

Choose Spot VMs for low priority workloads

Use managed services

# Cost Tradeoffs

As users design the workload, they must consider the tradeoffs between cost optimization and design aspects, such as security, scalability, resilience, and operability.

## Cost vs. reliability

The cost of high-availability components may exceed the acceptable downtime. If it exceeds the downtime cost, users can save by using Azure platform-managed replication and recovering data from the backup storage.

## Cost vs. performance efficiency

Factors impacting performance:

- Fixed or consumption-based provisioning
- Azure regions
- Caching
- Batch or real-time processing

# Cost Tradeoffs

An optimal design is not synonymous with a low-cost design. Low cost may incur additional risk.

## Cost vs. security

An increase in the security of the workload will increase costs.

## Cost vs. operation excellence

Investment in system monitoring and automation might increase the cost initially but will reduce the cost over time.



# Provision with Optimization

Best practices for optimizing spending include:

Eliminate waste

Leverage Azure Advisor

Choose appropriate VM sizes

Use purchase discounts



# Cost Savings with Azure Dev/Test Pricing

Azure Dev/Test pricing offers discounts up to 57% for typical web app dev/test environments running SQL Database and App Service.

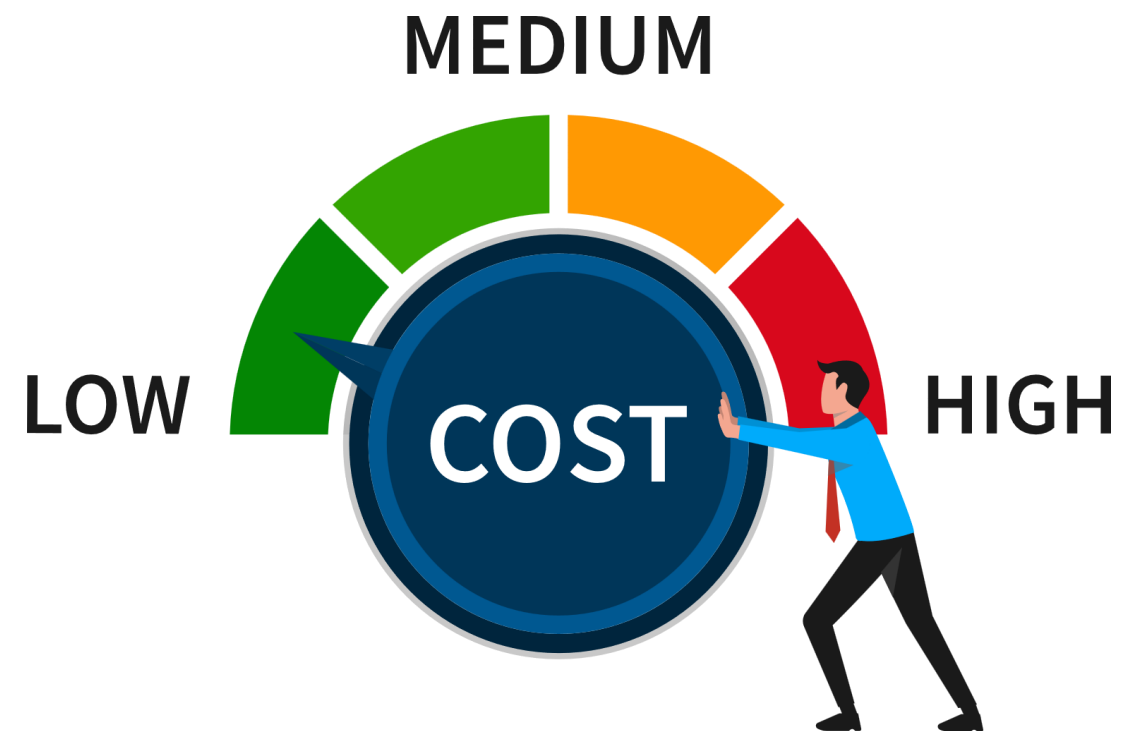


It provides discounted rates on various Azure services, including VMs, SQL Database, Logic Apps, App Service, Cloud Services, and HDInsight instances.

Source: <https://azure.microsoft.com/en-us/pricing/offers/dev-test/#overview>

# Azure Advisor Cost Recommendations

They can be viewed on the cost tab on Azure Advisor.



Azure Advisor helps to:

- Optimize spending on VMs or virtual machine scale sets (VMSS) by resizing or shutting down instances that are underutilized.
- Shutdown recommendations for the resources that have remained unused for the past 7 days.
- Resize SKU recommendations
- Give burstable recommendations

# Generating Report on Spend



**Duration: 10 Min.**

## **Problem Statement:**

As an Azure Architect, you've been asked to provide your company with an Azure cost solution to track cloud spending and make recommendations.

## **Outcome:**

You will be able to provide a comprehensive Azure cost solution to track cloud spending and make optimization recommendations.

**Note:** Refer to the demo document for detailed steps:  
01\_Generating\_Report\_on\_Spend

ASSISTED PRACTICE

# Assisted Practice: Guidelines

**Steps to be followed are:**

1. Generate a report on spend



# Creating Azure Billing Alerts



**Duration: 10 Min.**

## Problem Statement:

As an Azure Architect, you've been asked to provide your organization with an Azure cost solution to inform you when expenditure, based on usage or cost, hits or exceeds the amount indicated in the budget's alert condition.

## Outcome:

You will be able to set up Azure cost alerts to notify your organization when expenditure hits or exceeds budget thresholds.

**Note:** Refer to the demo document for detailed steps:  
02\_Creating\_Azure\_Billing\_Alerts

ASSISTED PRACTICE



# Assisted Practice: Guidelines

**Steps to be followed are:**

1. Create Azure billing alerts



## Key Takeaways

- Cost management and billing is a set of tools that allows the user to evaluate, manage, and optimize the costs of the workloads.
- Organizations must align finance, managers, and app teams to manage costs successfully.
- Azure provides tools to assist users in the assessment process before deploying cloud resources.
- After purchasing a reservation, it is important to track utilization.





## Knowledge Check

## Knowledge Check

01

**What are the two costs included in reservation charges?**

- A. Authorized cost and actual cost
- B. Actual cost and amortized cost
- C. Amortized cost and total cost
- D. Total cost and marginal cost



Knowledge  
Check

01

What are the two costs included in reservation charges?

- A. Authorized cost and actual cost
- B. Actual cost and amortized cost
- C. Amortized cost and total cost
- D. Total cost and marginal cost



The correct answer is **B**

**The reservation charges are shown as actual or amortized cost.**

Which of the following is NOT true about Azure cost management?

- A. Charges the users based on the consumption
- B. Shows usage-based costs consumed by Azure services
- C. Shows usage-based costs consumed by third-party marketplace offerings
- D. None of the above





Which of the following is NOT true about Azure cost management?

- A. Charges the users based on the consumption
- B. Shows usage-based costs consumed by Azure services
- C. Shows usage-based costs consumed by third-party marketplace offerings
- D. None of the above



The correct answer is **D**

**Azure cost management shows usage-based costs consumed by Azure services and third-party marketplace offerings and charges the users based on the consumption.**

**Knowledge  
Check**  
**03**

**Which of the following stakeholders are important to manage the costs successfully?**

- A. Organizational managers, finance, and app teams
- B. Customers, investors, and organizational managers
- C. App teams, customers, and finance
- D. App teams, investors, and finance



Knowledge  
Check  
03

Which of the following stakeholders are important to manage the costs successfully?

- A. Organizational managers, finance, and app teams
- B. Customers, investors, and organizational managers
- C. App teams, customers, and finance
- D. App teams, investors, and finance



The correct answer is **A**

**Organizations must align finance, managers, and app teams to manage costs successfully.**

**What are the items that a user must assess before deploying cloud resources?**

- A. The resources user plan to use
- B. The estimation of their cost
- C. The Azure offers that best meets user's needs
- D. All of the above



## Knowledge Check

04

What are the items that a user must assess before deploying cloud resources?

- A. The resources user plan to use
- B. The estimation of their cost
- C. The Azure offers that best meets user's needs
- D. All of the above



The correct answer is **D**

**Before deploying cloud resources, user must assess the following items:**

**The Azure offers that best meets user's needs, the resources user plan to use, and the estimation of their cost**

# TECHNOLOGY

**Thank You**